

Bootstrapping Multi-Modal Semantic-Vector Spaces

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1. Abstract

This study examines the impact of a shared Semantic-Vector space on Natural Language Understanding(NLU) tasks. We propose an alignment and evaluation architecture that builds on the work of [7] by providing a clear methodology for training and evaluating models that transition data into and out of a Semantic-Vector modality. This architecture allows us to examine Encoder and Decoder modules in isolation, and evaluate performance in semantic-vector translation operations.

Using open-weight models from vision, text and audio modalities we demonstrate how to rapidly bootstrap multimodal inference pipelines and provide insights into the efficacy of embedding-space alignment. This work highlights the potential of Semantic-Vector Modality to improve performance and efficiency of Deep Learning systems while reducing training and inference costs.

This research contributes to the growing corpus of work regarding dense Semantic-Vector Space(SVS) as a foundation for robust and generalizable AI systems.

Keywords

Natural Language Understanding, Large Concept Model, Semantic-Vector Space, Cross-Modal Alignment, Backtranslation

2. Introduction

Natural Language Understanding(NLU) has long been a central challenge in artificial intelligence. Traditional NLP systems rely on token-based representations, which, while effective, often fail to capture deeper semantic relationships[5]. Recent advances in representation learning suggest that high dimensional Semantic-Vector spaces may provide a more robust foundation for cross-modal

understanding[9] as well enable more efficient training[7] and computation[7][8][9].

Using Semantic-Vector space as a common operational modality for inference tasks shifts the responsibility of consuming or producing data in a given modality to an independent Encoder or Decoder module. Additionally, removing modality dependent computational features from inference modules has been shown to significantly impact performance for textual translation[8] and reasoning tasks[7][9].

This project proposes an architecture and tooling focused on aligning open-weight models to a shared Semantic-Vector modality. We use concepts grounded in Backtranslation to allow independent alignment and evaluation of Encoder and Decoder modules across text and image modalities.

We propose to advance foundational works [7] and [8] in two ways; by 1) providing a simplified pathway for module alignment to the Semantic-Vector Space; and 2) allowing for cross-modality alignment. We show that given an initial Embedding Module, it is possible to align an Encoder or Decoder of any modality to the shared Semantic-Vector space. We also show that Semantic-Vector aligned models out-perform their Text modality counterparts to confirm the utility of the Semantic-Vector modality.

3. Related Concepts

Our background research into Spoken Language Understanding and Natural Language Processing uncovered a number of concepts and processes that serve as the foundation of this project.

Semantic-Vector Space

Every generative LLM utilizes an internal latent space for processing, taking input of a given

modality and encoding it to an internal vector-space representation before any true inference tasks begin. Models then translate inference results out of this vector space to generate final outputs. These repeated translations into and out of different semantic spaces are inherently prone to semantic drift and translation errors[7][1][2][Any paper talking about the pitfalls of cascade architectures].

The architecture proposed in [7] removes the need for model-specific semantic encoding by training inference models to a shared, dense, Semantic-Vector space(Fig. 1). Researchers on the project found a consistent improvement in inference tasks, decreased token usage, and lower semantic drift.

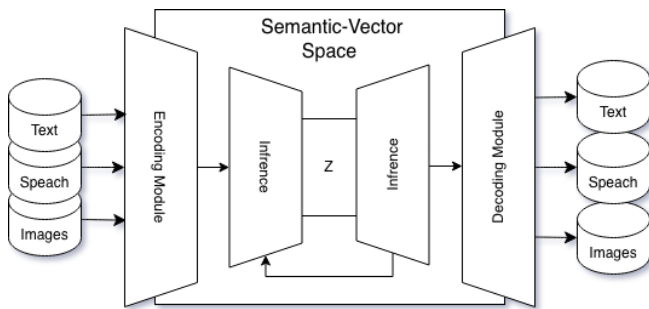


Fig 1.

Aligning inference resources to a common Semantic-Vector modality additionally creates clear separations of concern between Inference, Encoding, and Decoding modules. This architectural arrangement allows us to train and fine-tune modules in isolation, simplifying training processes, and allowing for scientifically rigorous evaluations of module performance.

Backtranslation

Backtranslation is a strategy used for Data Augmentation that uses known translation pairings (A, B) and (B, C), to generate synthetic pairings of (A, C)[2]. This means that given known pairings of (Image, Text) and (Text, Semantic-Vector), we can

synthesize a training target of (Image, Semantic-Vector). It is important to note that the quality of the synthetic relationship (A,C) is dependent on the semantic fidelity of (A,B) labeling and (B,C) encoding.[2]

Cross Modal Alignment

[1][3][5][4] Examined Cross Modal alignment in pursuit of End to End Spoken Language understanding. All discuss strategies, primarily Contrastive Learning, for cross modal alignment. Aligning models to a common embedding space resulted in embeddings with increased fidelity[4].

Contrastive Learning

Contrastive Learning is a semi-supervised learning algorithm that trains models to group similar things together and dissimilar things further apart[5]. The strategy relies on Contrastive Triplets consisting of a) Query point b) Positive(Similar) Sample and c) Negative(Dissimilar) sample

Distillation Learning

Distillation learning is a training strategy that transfers knowledge from one model to another by using the Teacher model's output as ground truth when calculating loss. [1] shows how using momentum distillation enables models to learn from imperfectly labeled data due to cross modality translation errors.

Joint Representation Learning

This involves learning a unified representation from multiple sources of data, such as images and text. It uses encoder models to create a combined, shared, representation space from multimodal data.[1]

4. Methodology

This work's primary contribution is an Encoder and Decoder evaluation and alignment pipeline that allows for the iterative adoption of new input and

output modalities to a common Semantic-Vector space.

Encoder and Decoder Architecture

Inspired by [11] Encoders and Decoders follow a BERT architecture consisting of a pretrained encoder/decoder module and a Multi-Layer Preceptron(MLP), as seen in Fig 2. Using a simple MLP as a translational module between the Semantic-Vector space and the Pretrained Coder’s modality allows us to quickly align modules.

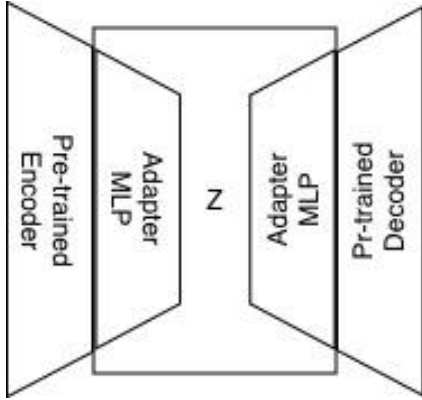


Fig. 2

This configuration was shown by [11] to be highly performant while being less prone to overfitting, two orders of magnitude less costly to train, and more robust to distribution shift.[11] This gain in efficiency is due to the fact that fine-tuning is focused on MLP Adapter instead of the large encoder module.

Encoder Alignment

Encoder alignment is driven by the momentum cross modal loss calculations laid out in [1], relying on cross-modal momentum-based distillation to overcome the imperfect nature of cross-modality labeling to jointly learn the shared Semantic-Vector space. Our architecture, seen in Fig 3, Uses cross-modality paired data to align encoder modules to a shared vector space. See Appendix A for specific loss calculation.

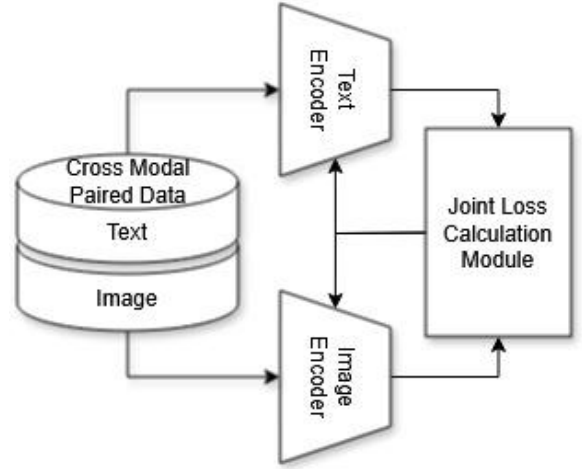


Fig. 3

Decoder Alignment

Decoder alignment architecture is shown in Fig. 4. It is important to note that to train a given Decoder, an Encoder of the same modality must already be available. Decoder Loss is evaluated on two metrics: Reconstruction fidelity, and Semantic-Vector Loss after output is re-encoded to the shared vector space. Both measure overall reconstruction quality in different ways. Vector-Space loss measures the semantic drift introduced during Decoding and Encoding processes, while Reconstruction Loss measures the decoder’s fidelity to training data’s aesthetic features.

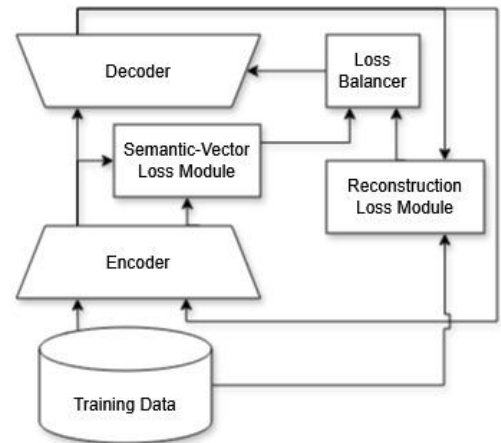


Fig. 4

Evaluation

Our easiest task is to show low contrastive-learning error on unseen data. And showing that semantically-linked multi-modal data is encoded to similar vectors.

Align Text, and Image Encoder and Decoder modules to show cross-modal utility. Also, the other papers only did Text and Voice speech, Images would be a significant contribution.

All in we will be training 4 modules: (these models may change as implementation concerns arise).

Encoders: (think Modality -> Text)

- 1) FacebookBART - Text
- 2) Microsoft/Kosmos-2 - Image
- 3) Whisper - Spoken Language
- 4) Vit-gpt2-image-captionin - Static Images

Decoders: (think Text -> Modality)

- 1) MiniGPT
- 2) Stable-Diffusion

Datasets:

[MultiBench](#) is a comprehensive evaluation and set of implementations of existing multimodal machine learning algorithms. It contains 20 methods spanning different methodological innovations in (1) data preprocessing, (2) fusion paradigms, (3) optimization objectives, and (4) training procedures.

[MultiBench Github](#)

[Conceptual 12M](#) a dataset with ~12 million image-text pairs meant to be used for vision-and-language pre-training.

5. Results and Discussion

Specific Test results and ablative studies.

Discuss our interpretations of the results, what do they mean? Is there a particular architecture that works best in encoder/decoders?

Conclusion and Future Work

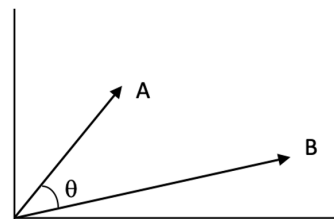
Summarize Findings and give brief interpretations. If someone just read the abstract and this section, they should be able to get the general topic and outcomes of the paper.

This work enables further research into the properties of semantic-vector spaces. Next steps in research will pursue both statistical analysis of vector-space relationships and agentic flows within the semantic-vector modality.

7. Apendex

A: Equations

Cosine Similarity: drives similarity of two vectors by the Cosine of the angle between the vectors. [10][Images from same source]



$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

Cross Modal Momentum Join Loss:

balances loss between multiple models to jointly inform a shared embedding space[1]
[EQUATION AND VISUAL HERE]

B: Related Works Descriptions

[1] Addresses data scarcity in SLU by aligning speech and text embeddings.

[2] Proposes using speech synthesis to generate large artificial datasets, reducing reliance on recorded in-domain speech.

[3] Combines deep architectures with data augmentation. This approach achieves results without ASR targets or pretrained models.

[4] Treats SLU as a multi-modal task by aligning audio and text embeddings in a shared latent space.

[5] Introduces a speech-text embedding model that bypasses ASR, mapping both modalities into a shared space.

[6] Combines BERT's contextual modeling with BART's generative ability to produce concise, high-quality summaries.

[7] Trains sequence-to-sequence models for auto-regressive prediction using approaches like MSE regression, diffusion-based generation, and quantized SONAR embeddings.

[8] Introduces a fixed-size embedding space with a single text encoder covering 200 languages and speech encoders for 57 languages.

[9] Introduces a latent reasoning paradigm where the model reuses hidden states as "continuous thoughts" instead of decoding into text.

[10] Cosine similarity quantifies the similarity between two vectors by measuring the cosine of the angle between them.

[11] Leverages pretrained unimodal encoders with curated alignment data to reduce training cost and time.

C: Resources

The code for this project is publicly available at: <https://github.com/WatsonWBlair/cs627>

MultiBench is available at <https://github.com/pliang279/MultiBench> and <https://cmu-multicomp-lab.github.io/multiben/ch/>

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